$$\langle x_1, \ldots, x_n \rangle \preceq_{\langle\!\langle \mathbf{P}_1, \ldots, \mathbf{P}_n \rangle\!\rangle} \langle y_1, \ldots, y_n \rangle$$

$$x_i \preceq_{\mathbf{P}_i} y_i \text{ for all } i \in \{1, \ldots, n\}$$